

CHAPTER 10

ICE LOADS—ATMOSPHERIC ICING

10.1 GENERAL

Atmospheric ice loads caused by freezing rain, snow, and in-cloud icing shall be considered in the design of ice-sensitive structures. In areas where records or experience indicate that snow or in-cloud icing produces larger loads than freezing rain, site-specific studies shall be used. Structural loads caused by hoarfrost are not a design consideration. Roof snow loads are covered in Chapter 7.

10.1.1 Site-Specific Studies. Mountainous terrain and gorges shall be examined for unusual icing conditions. Site-specific studies shall be used to determine the 500-year mean recurrence interval ice thickness, concurrent wind speed, and concurrent temperature in

1. Alaska;
2. areas where records or experience indicate that snow or in-cloud icing produces larger loads than freezing rain;
3. special icing regions shown in Figs. 10.4-2, 10.4-4, and 10.4-5; and
4. mountainous terrain and gorges where examination indicates unusual icing conditions exist.

Site-specific studies shall be subject to review and approval by the Authority Having Jurisdiction.

In lieu of using the mapped values, it shall be permitted to determine the ice thickness, the concurrent wind speed, and the concurrent temperature for a structure from local meteorological data based on a 500-year mean recurrence interval provided that

1. The quality of the data for wind and type and amount of precipitation has been taken into account.
2. A robust ice accretion algorithm has been used to estimate uniform ice thicknesses and concurrent wind speeds from these data.
3. Extreme-value statistical analysis procedures acceptable to the Authority Having Jurisdiction have been used in analyzing the ice thickness and concurrent wind speed data.
4. The length of record and sampling error have been taken into account.

10.1.2 Dynamic Loads. Dynamic loads, such as those resulting from galloping, ice shedding, and aeolian vibrations, that are caused or enhanced by an ice accretion on a flexible structural member, component, or appurtenance are not covered in this section.

10.1.3 Exclusions. Electric transmission systems, communications towers and masts, and other structures for which national standards exist are excluded from the requirements of this section. Applicable standards and guidelines include the NES, ASCE Manual 74, and ANSI/EIA/TIA-222.

10.2 DEFINITIONS

The following definitions apply only to the provisions of this chapter.

COMPONENTS AND APPURTENANCES: Nonstructural elements that may be exposed to atmospheric icing. Examples are ladders, handrails, antennas, waveguides, radio frequency (RF) transmission lines, pipes, electrical conduits, and cable trays.

FREEZING RAIN: Rain or drizzle that falls into a layer of subfreezing air at the Earth's surface and freezes on contact with the ground or an object to form glaze ice.

GLAZE: Clear, high-density ice.

HOARFROST: An accumulation of ice crystals formed by direct deposition of water vapor from the air onto an object.

ICE-SENSITIVE STRUCTURES: Structures for which the effect of an atmospheric icing load governs the design of part or all of the structure. This effect includes, but is not limited to, lattice structures, guyed masts, overhead lines, light suspension and cable-stayed bridges, aerial cable systems (e.g., for ski lifts and logging operations), amusement rides, open catwalks and platforms, flagpoles, and signs.

IN-CLOUD ICING: Icing that occurs when supercooled cloud or fog droplets carried by the wind freeze on impact with objects. In-cloud icing usually forms rime but may also form glaze.

RIME: White or opaque ice with entrapped air.

SNOW: Snow that adheres to objects by some combination of capillary forces, freezing, and sintering.

10.3 SYMBOLS

A_i = cross-sectional area of ice

A_s = surface area of one side of a flat plate or the projected area of complex shapes

D = diameter of a circular structure or member as defined in Chapter 29, in ft (m)

D_c = diameter of the cylinder circumscribing an object

f_z = factor to account for the increase in ice thickness with height

I_i = importance factor for ice thickness from Table 1.5-2 based on the Risk Category from Table 1.5-1

I_w = importance factor for concurrent wind pressure from Table 1.5-2 based on the Risk Category from Table 1.5-1

K_{zt} = topographic factor as defined in Chapter 26

q_z = velocity pressure evaluated at height z above ground, in lb/ft² (N/m²) as defined in Chapter 29

r = radius of the maximum cross section of a dome or radius of a sphere

t = nominal ice thickness on a cylinder caused by freezing rain at a height of 33 ft (10 m) from Figs. 10.4-2 through 10.4-6 in inches (mm)

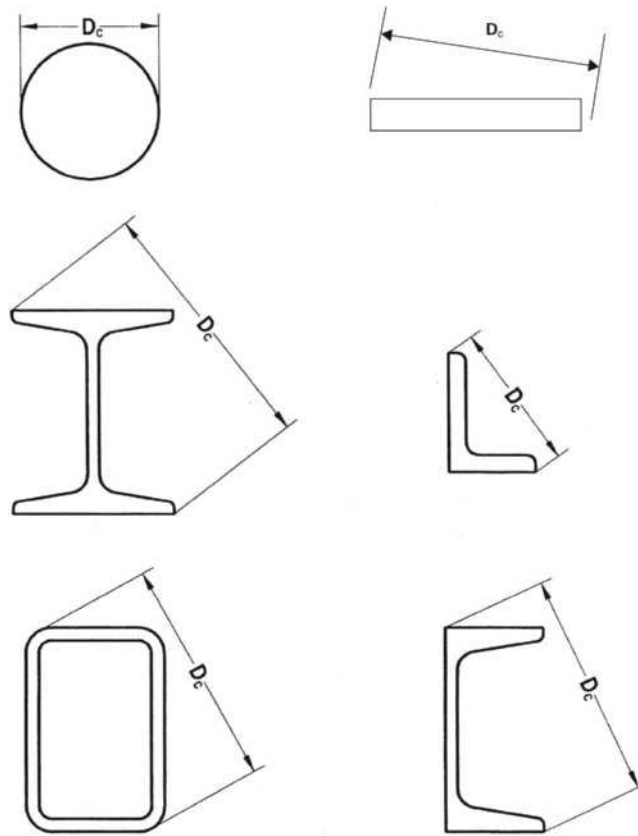


FIGURE 10.4-1 Characteristic Dimension D_c for Calculating the Ice Area for a Variety of Cross-Sectional Shapes

t_d = design ice thickness in inches (mm) from Eq. (10.4-5)
 V_c = concurrent wind speed in mph (m/s) from Figs. 10.4-2 through 10.4-6
 V_i = volume of ice
 z = height above ground in ft (m)
 ϵ = solidity ratio as defined in Chapter 29

10.4 ICE LOADS CAUSED BY FREEZING RAIN

10.4.1 Ice Weight. The ice load shall be determined using the weight of glaze ice formed on all exposed surfaces of structural members, guys, components, appurtenances, and cable systems. On structural shapes, prismatic members, and other similar shapes, the cross-sectional area of ice shall be determined by

$$A_i = \pi t_d (D_c + t_d) \quad (10.4-1)$$

D_c is shown for a variety of cross-sectional shapes in Fig. 10.4-1.

On flat plates and large three-dimensional objects such as domes and spheres, the volume of ice shall be determined by

$$V_i = \pi t_d A_s \quad (10.4-2)$$

For a flat plate, A_s shall be the area of one side of the plate; for domes and spheres, A_s shall be determined by

$$A_s = \pi r^2 \quad (10.4-3)$$

It is acceptable to multiply V_i by 0.8 for vertical plates and 0.6 for horizontal plates.

The ice density shall be not less than 56 pcf (900 kg/m³).

10.4.2 Nominal Ice Thickness. Figs. 10.4-2 through 10.4-6 show the equivalent uniform radial thicknesses t of ice caused by freezing rain at a height of 33 ft (10 m) over the contiguous 48 states and Alaska for a 500-year mean recurrence interval. Also shown are concurrent 3-s gust wind speeds. Thicknesses for Hawaii, and for ice accretions caused by other sources in all regions, shall be obtained from local meteorological studies.

10.4.3 Height Factor. The height factor f_z used to increase the radial thickness of ice for height above ground z shall be determined by

$$f_z = \left(\frac{z}{33} \right)^{0.10} \quad \text{for } 0 \text{ ft} < z \leq 900 \text{ ft} \quad (10.4-4)$$

where $f_z = 1.4$ for $z > 900$ ft.

$$f_z = \left(\frac{z}{10} \right)^{0.10} \quad \text{for } 0 \text{ m} < z \leq 275 \text{ m} \quad (10.4-4\text{si})$$

where $f_z = 1.4$ for $z > 275$ m.

10.4.4 Importance Factors. Importance factors to be applied to the radial ice thickness and wind pressure shall be determined from Table 1.5-2 based on the Risk Category from Table 1.5-1. The importance factor I_i shall be applied to the ice thickness, not the ice weight, because the ice weight is not a linear function of thickness.

10.4.5 Topographic Factor. Both the ice thickness and concurrent wind speed for structures on hills, ridges, and escarpments are higher than those on level terrain because of

wind speed-up effects. The topographic factor for the concurrent wind pressure is K_{zt} , and the topographic factor for ice thickness is $(K_{zt})^{0.35}$, where K_{zt} is obtained from Eq. (26.8-1).

10.4.6 Design Ice Thickness for Freezing Rain. The design ice thickness t_d shall be calculated from Eq. (10.4-5).

$$t_d = t I f_z (K_{zt})^{0.35} \quad (10.4-5)$$

10.5 WIND ON ICE-COVERED STRUCTURES

Ice accreted on structural members, components, and appurtenances increases the projected area of the structure exposed to wind. The projected area shall be increased by adding t_d to all free edges of the projected area. Wind loads on this increased projected area shall be used in the design of ice-sensitive structures. Figs. 10.4-2 to 10.4-6 include 3-s gust wind speeds at 33 ft (10 m) above grade that are concurrent with the ice loads caused by freezing rain. Wind loads shall be calculated in accordance with Chapters 26 through 31 as modified by Sections 10.5.1 through 10.5.5.

10.5.1 Wind on Ice-Covered Chimneys, Tanks, and Similar Structures. Force coefficients C_f for structures with square, hexagonal, and octagonal cross sections shall be as given in Fig. 29.5-1. Force coefficients C_f for structures with round cross sections shall be as given in Fig. 29.4-1 for round cross sections with $D\sqrt{q_z} \leq 2.5$ for all ice thicknesses, wind speeds, and structure diameters.

10.5.2 Wind on Ice-Covered Solid Freestanding Walls and Solid Signs. Force coefficients C_f shall be as given in Fig. 29.3-1 based on the dimensions of the wall or sign, including ice.

10.5.3 Wind on Ice-Covered Open Signs and Lattice Frameworks. The solidity ratio ϵ shall be based on the projected area, including ice. The force coefficient C_f for the projected area of flat members shall be as given in Fig. 29.4-2. The force coefficient C_f for rounded members and for the additional projected area caused by ice on both flat and rounded members shall be as given in Fig. 29.4-2 for rounded members with $D\sqrt{q_z} \leq 2.5$ for all ice thicknesses, wind speeds, and member diameters.

10.5.4 Wind on Ice-Covered Trussed Towers. The solidity ratio ϵ shall be based on the projected area, including ice. The force coefficients C_f shall be as given in 29.4-3. It is acceptable to reduce the force coefficients C_f for the additional projected area caused by ice on both round and flat members by the factor for rounded members in Note 3 of 29.4-3.

10.5.5 Wind on Ice-Covered Guys and Cables. The force coefficient C_f (as defined in Chapter 29) for ice-covered guys and cables shall be 1.2.

10.6 DESIGN TEMPERATURES FOR FREEZING RAIN

The design temperatures for ice and wind-on-ice caused by freezing rain shall be either the temperature for the site shown in Figs. 10.6-1 and 10.6-2 or 32°F (0°C), whichever gives the maximum load effect. The temperature for Hawaii shall be 32°F

(0°C). For temperature-sensitive structures, the load shall include the effect of temperature change from everyday conditions to the design temperature for ice and wind on ice. These temperatures are to be used with ice thicknesses for all mean recurrence intervals. The design temperatures are considered to be concurrent with the design ice load and the concurrent wind load.

10.7 PARTIAL LOADING

The effects of a partial ice load shall be considered when this condition is critical for the type of structure under consideration. It is permitted to consider this to be a static load.

10.8 DESIGN PROCEDURE

1. The nominal ice thickness, t , the concurrent wind speed, V_c , and the concurrent temperature for the site shall be determined from Figs. 10.4-2 to 10.4-6 and 10.6-1 and 10.6-2 or a site-specific study.
2. The topographic factor for the site, K_{zt} , shall be determined in accordance with Section 10.4.5.
3. The importance factor for ice thickness, I_i , shall be determined in accordance with Section 10.4.4.
4. The height factor, f_z , shall be determined in accordance with Section 10.4.3 for each design segment of the structure.
5. The design ice thickness, t_d , shall be determined in accordance with Section 10.4.6, Eq. (10.4-5).
6. The weight of ice shall be calculated for the design ice thickness, t_d , in accordance with Section 10.4.1.
7. The velocity pressure, q_z , for wind speed, V_c , shall be determined in accordance with Section 29.3 using the importance factor for concurrent wind pressure, I_w , determined in accordance with Section 10.4.4.
8. The wind force coefficients, C_f , shall be determined in accordance with Section 10.5.
9. The gust effect factor shall be determined in accordance with Section 26.9.
10. The design wind force shall be determined in accordance with Chapter 29.
11. The iced structure shall be analyzed for the load combinations in either Section 2.3 or 2.4.

10.9 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

This section lists the consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

ASCE Manual 74, *Guidelines for Electrical Transmission Line Structural Loading*, American Society of Civil Engineers, 2009.

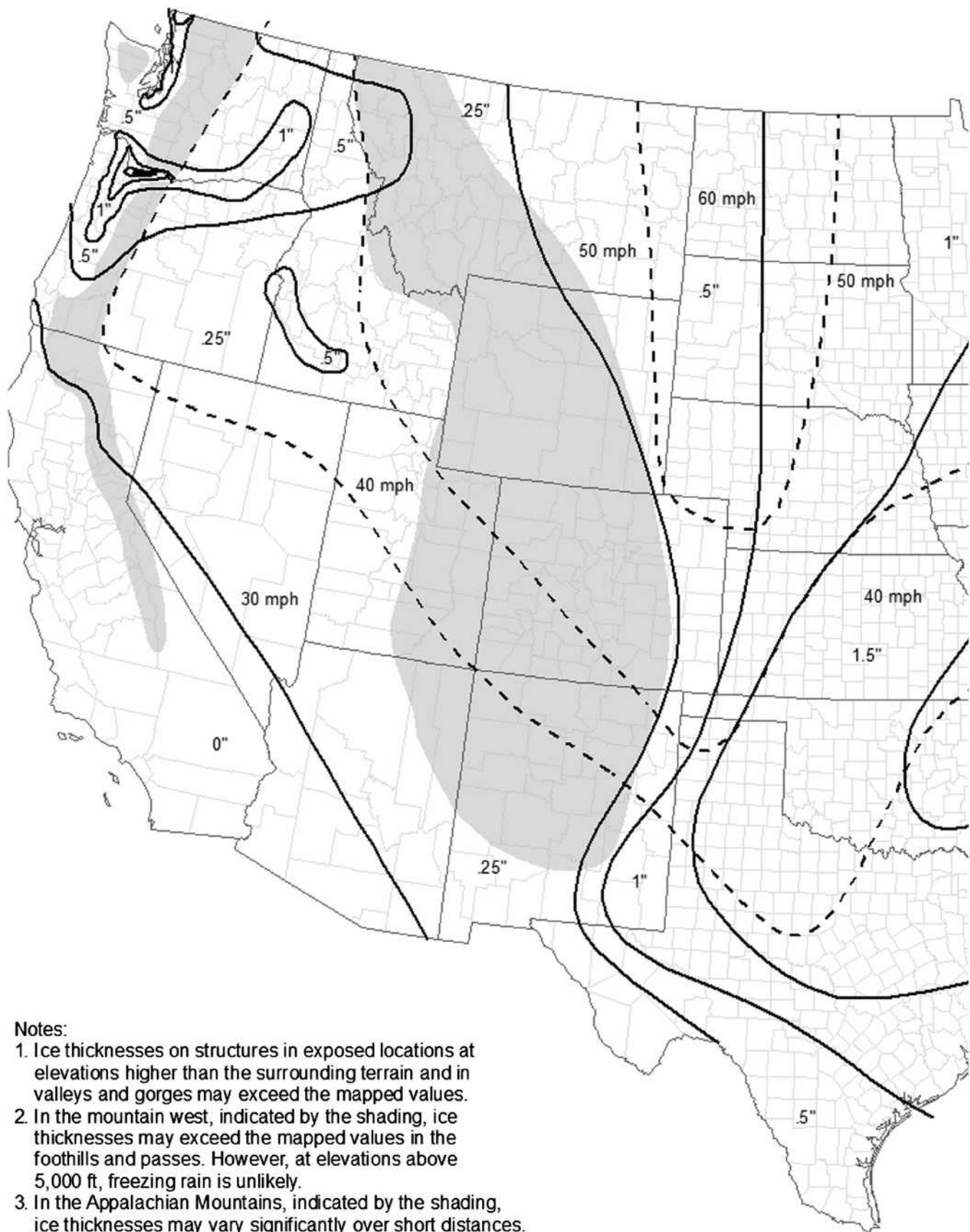
Cited in: Section 10.1.3

ANSI/EIA/TIA-222, *Structural Standards for Steel Antenna Towers and Antenna Supporting Structures*, American National Standards Institute, 2013.

Cited in: Section 10.1.3

NESC, *National Electrical Safety Code*, IEEE, 2012.

Cited in: Section 10.1.3



Notes:

1. Ice thicknesses on structures in exposed locations at elevations higher than the surrounding terrain and in valleys and gorges may exceed the mapped values.
2. In the mountain west, indicated by the shading, ice thicknesses may exceed the mapped values in the foothills and passes. However, at elevations above 5,000 ft, freezing rain is unlikely.
3. In the Appalachian Mountains, indicated by the shading, ice thicknesses may vary significantly over short distances.

FIGURE 10.4-2 Equivalent Radial Ice Thicknesses Caused by Freezing Rain with Concurrent 3-Second Gust Speeds, for a 500-Year Mean Recurrence Interval

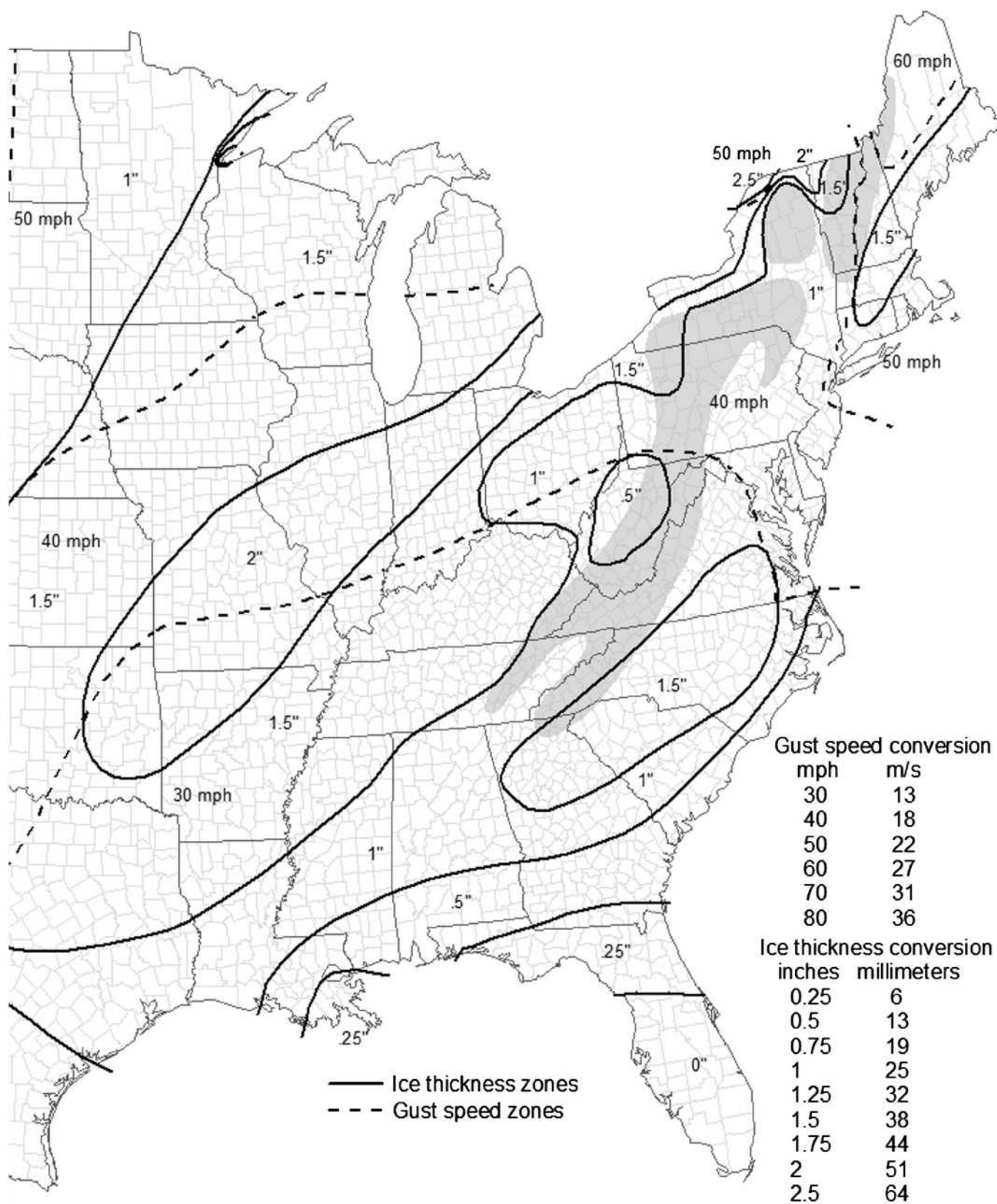


FIGURE 10.4-2 (Continued)

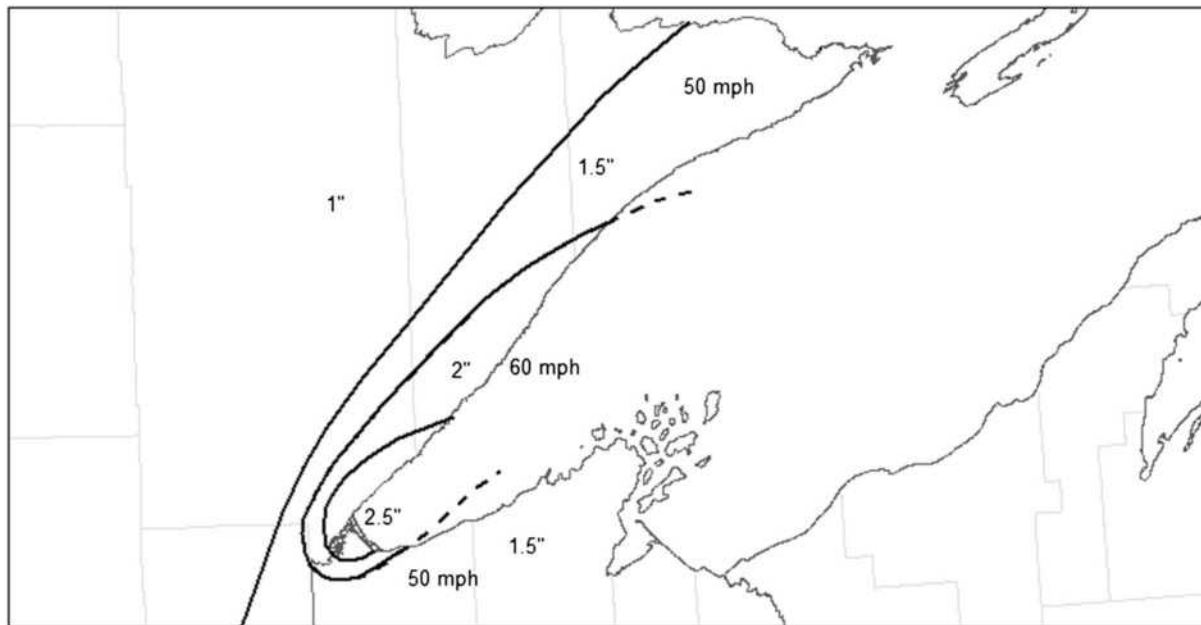


FIGURE 10.4-3 Lake Superior Detail

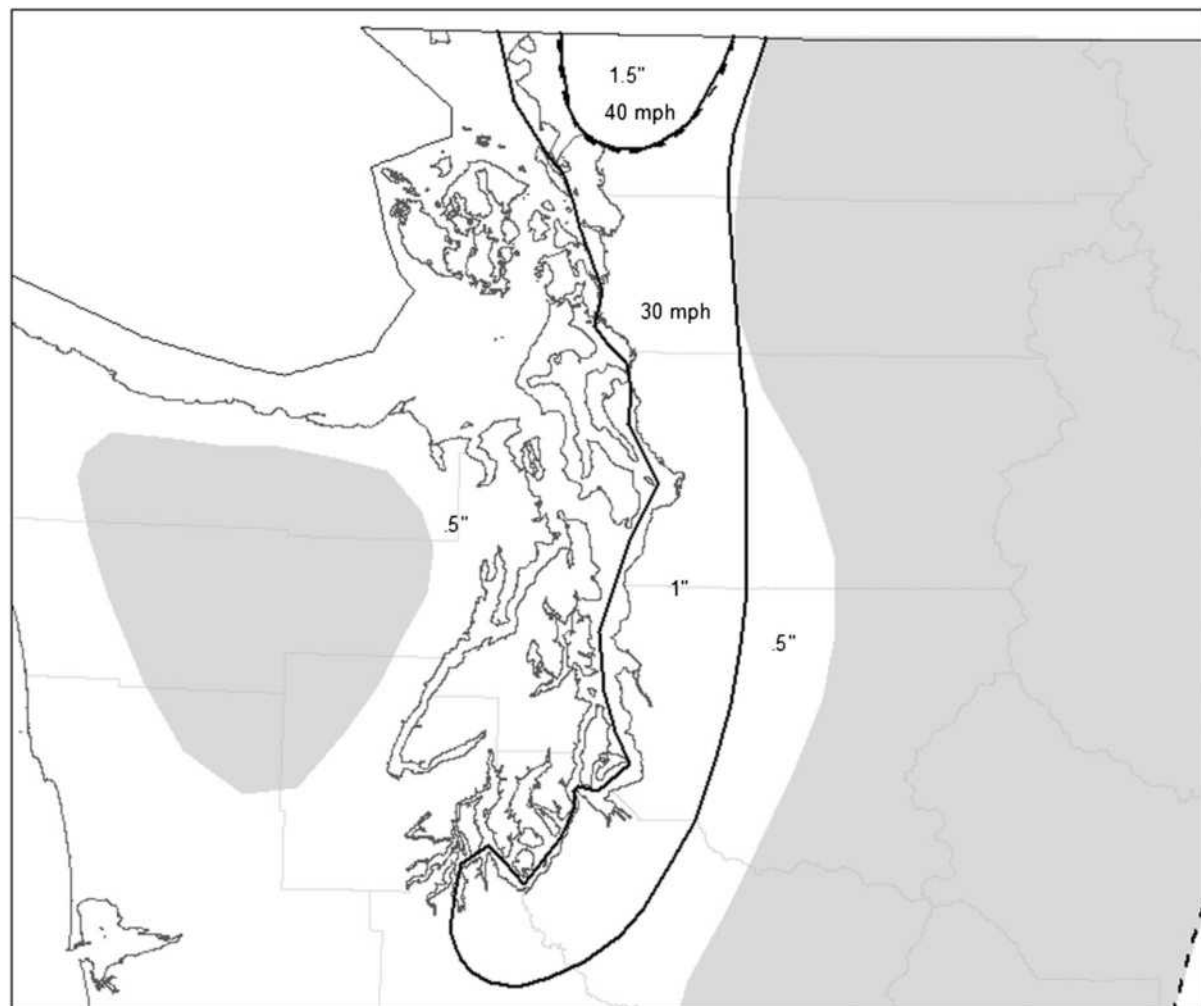


FIGURE 10.4-4 Fraser Valley Detail

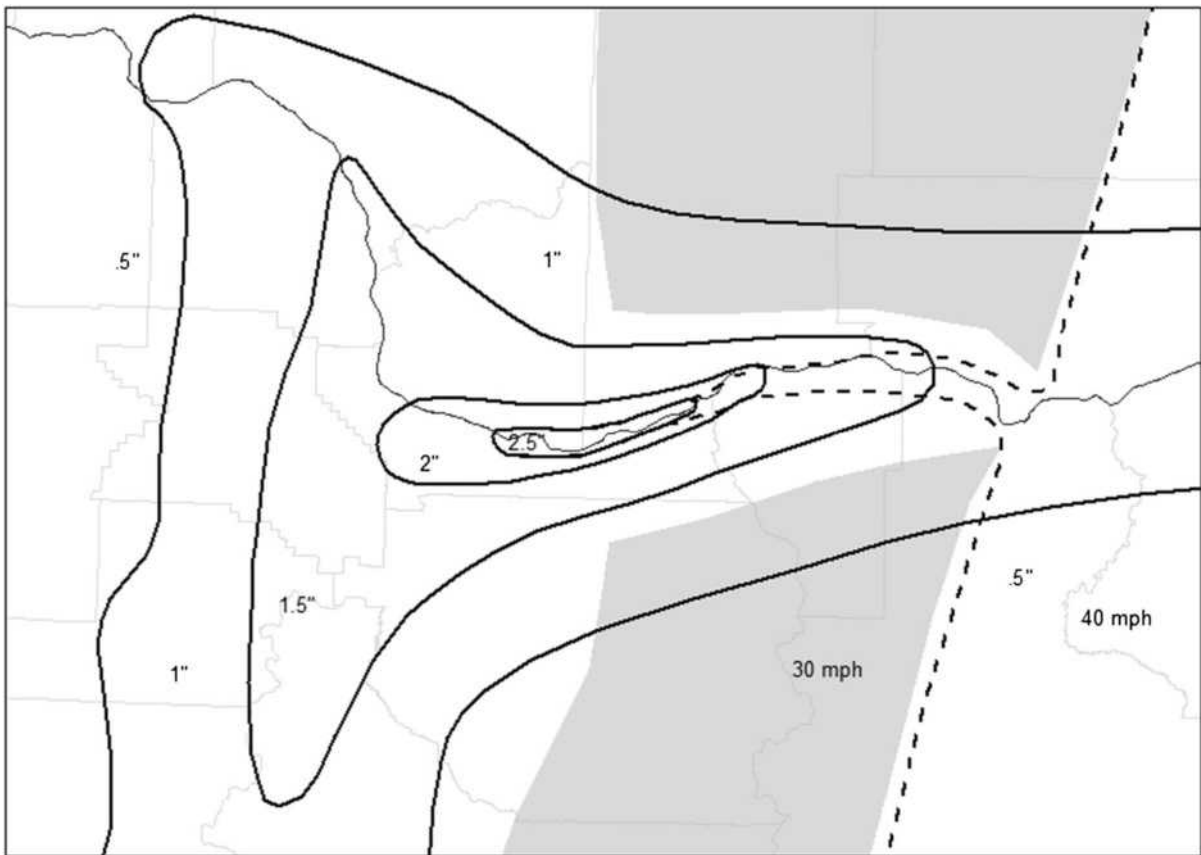


FIGURE 10.4-5 Columbia River Gorge Detail

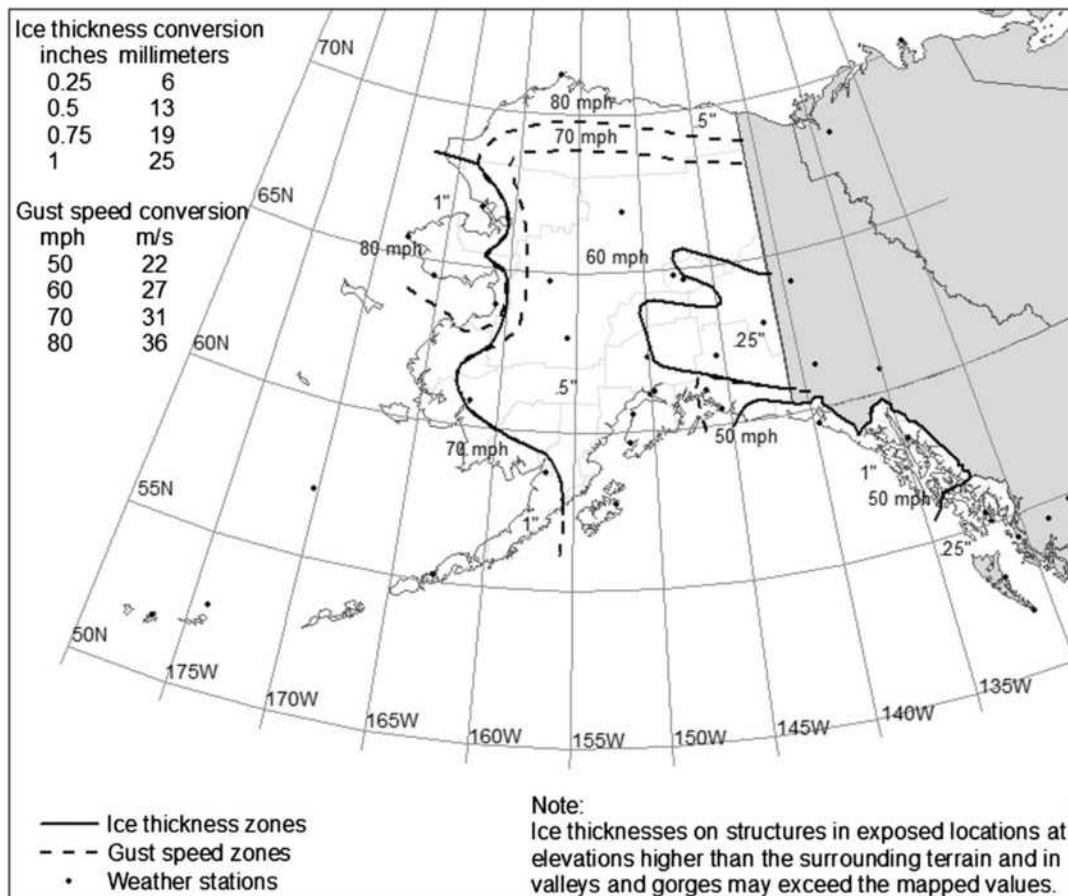


FIGURE 10.4-6 500-Year Mean Recurrence Interval Uniform Ice Thicknesses Caused by Freezing Rain with Concurrent 3-Second Gust Speeds: Alaska

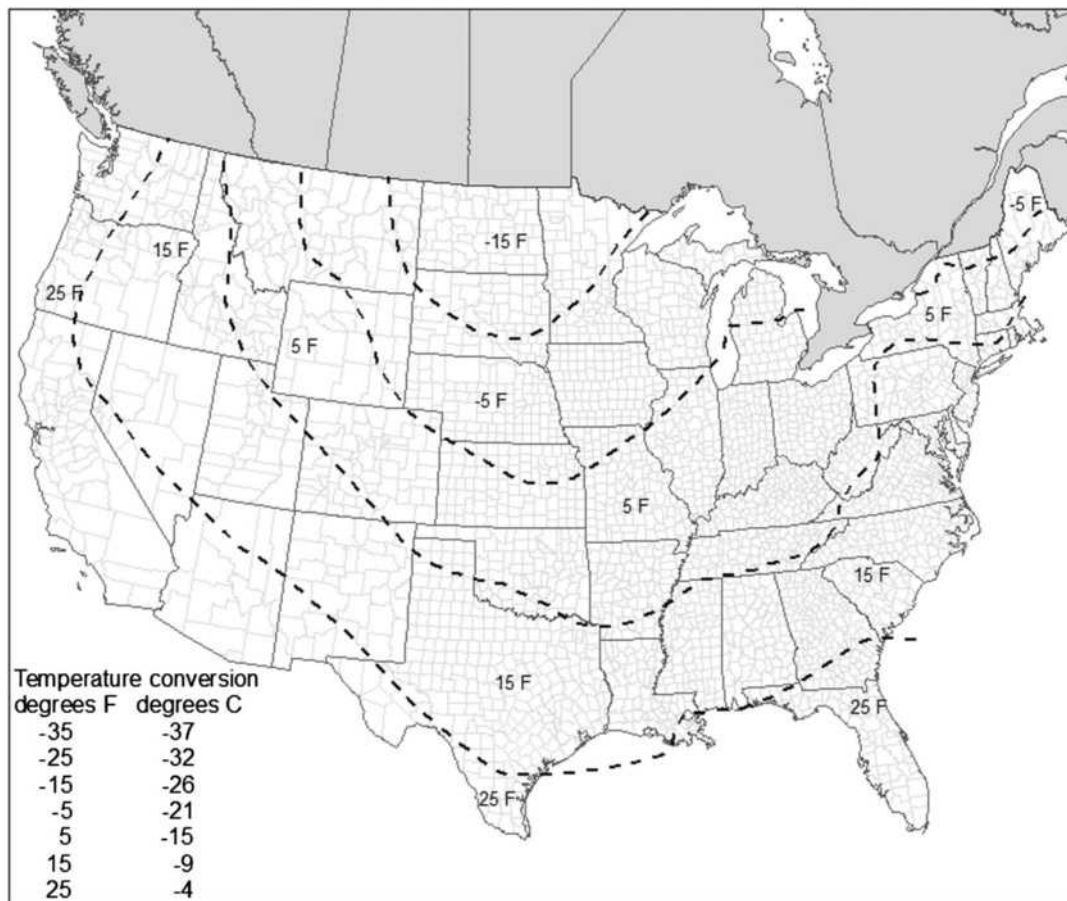


FIGURE 10.6-1 Temperatures Concurrent with Ice Thicknesses Caused by Freezing Rain: Contiguous 48 States

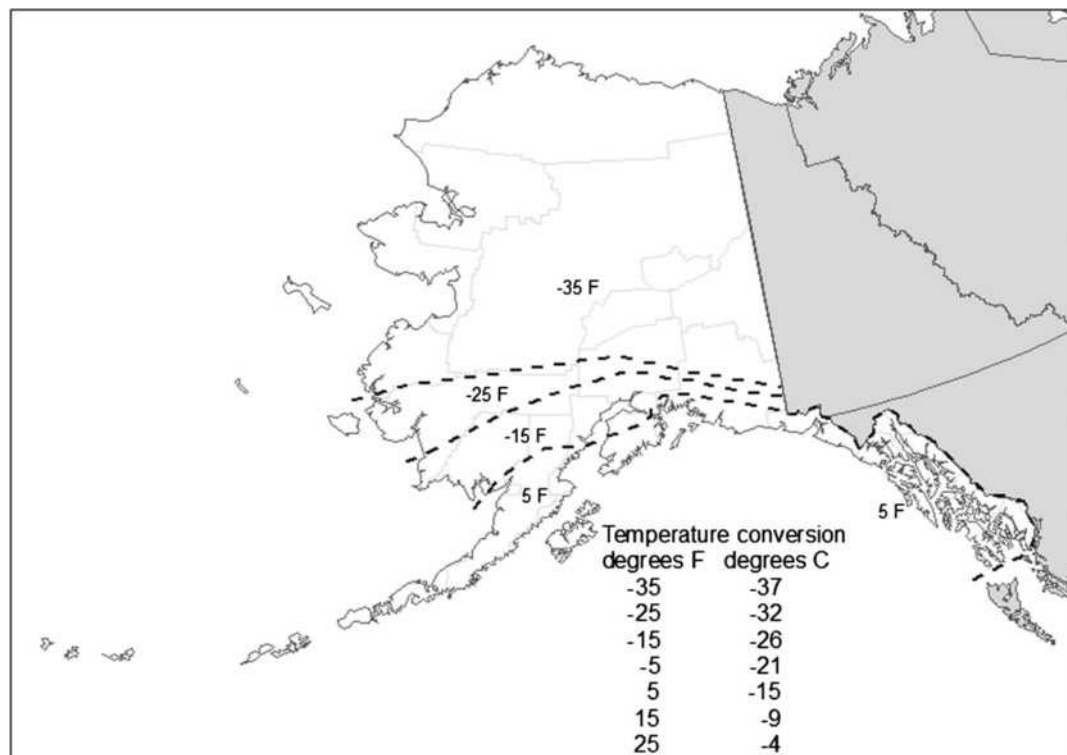


FIGURE 10.6-2 Temperatures Concurrent with Ice Thicknesses Caused by Freezing Rain: Alaska